



INTRODUCTION TO PROBABILITY

PROBABILITY FOR DATA SCIENCE

AIMS OF THE SESSION

- Be familiar with the **fundamental rules of probability**
- Learn how to compute particular **statistics** to describe data
- Be comfortable working with common probability distributions for modelling **discrete** and **continuous** random variables
- Cover the intuition and mathematical formulation of **Maximum Likelihood Estimation**

WHAT IS PROBABILITY THEORY?

Probability is the science of how **likely** an event is to happen: "*What is the probability that I will roll two sixes on the dice?*"



PROBABILITY OF AN EVENT: FORMAL DEFINITION

- The probability that an event A will happen is:

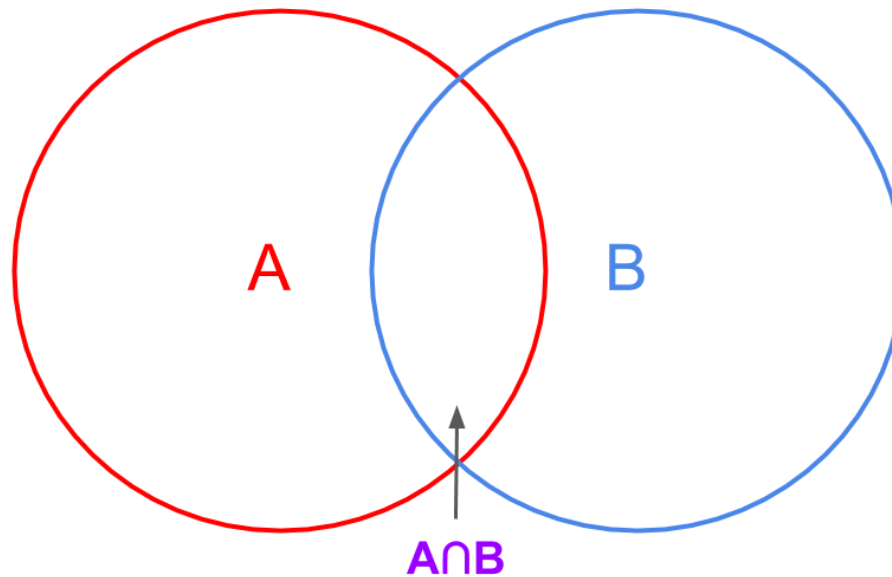
$$P(A) = \frac{\text{number of outcomes that lead to that event}}{\text{total number of possible outcomes}}$$

- Probabilities are always between 0 and 1
- The probability of the event not occurring is $1 - P(A)$
- Dice example:

Rolling a six on both dice is the only outcome which leads to our event, out of 36 total possible outcomes. Therefore the probability of rolling two sixes is 0.0278.

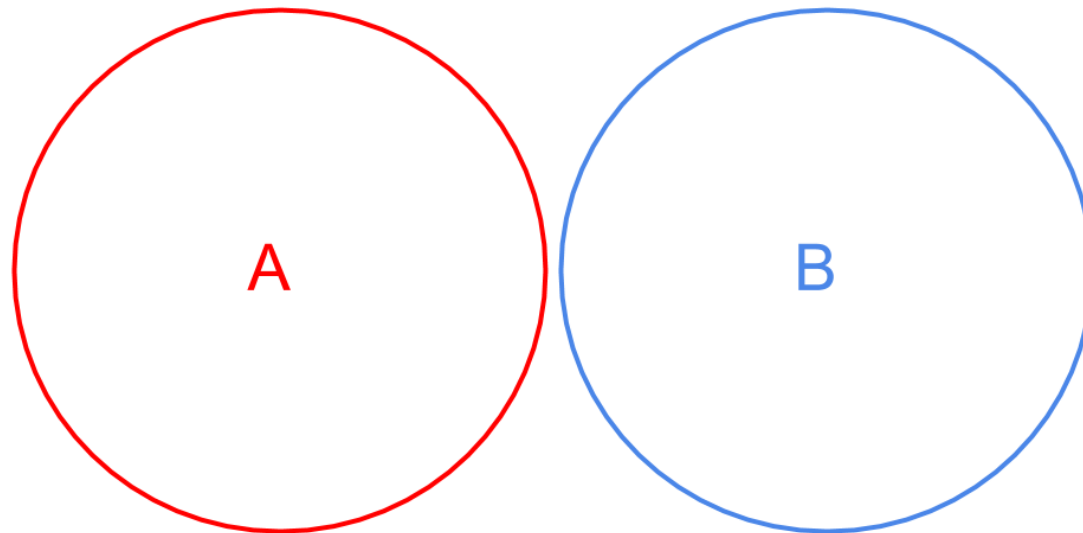
TWO EVENTS

- Probability of either event occurring (A or B) = $P(A \cup B)$
- Probability of both events occurring (A and B) = $P(A \cap B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



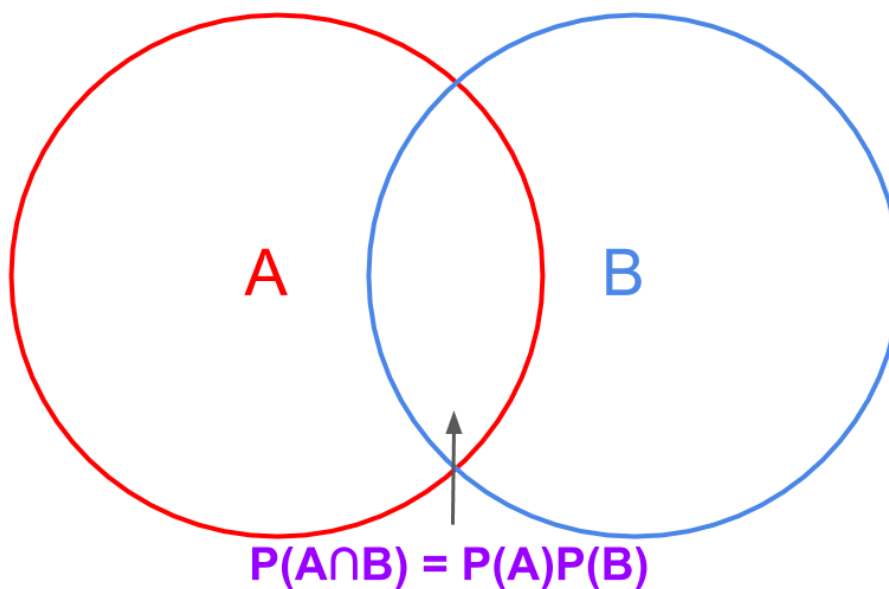
MUTUALLY EXCLUSIVE

- Two events are mutually exclusive if they cannot occur at the same time
 - $P(A \cap B) = 0$
- Example of mutually exclusive events:
 - A = the event of rolling two sixes on two dice
 - B = the event that the sum of the values on two dice is ten



INDEPENDENCE

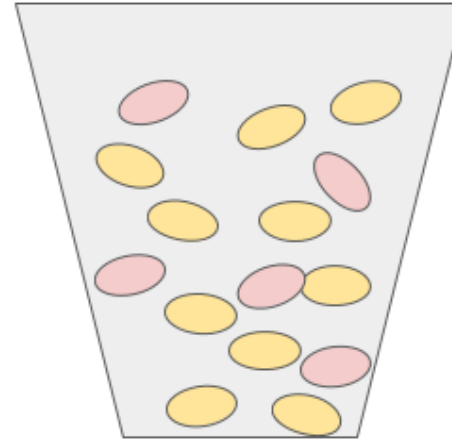
- The outcome of one event does not depend on the outcome of another
 - $P(A \cap B) = P(A)P(B)$
- Example of independent events:
 - A = the event that I cook pasta for dinner
 - B = the event that Chelsea win the Premier League



INDEPENDENT EVENTS: EXAMPLE

- Events:

- A = event that the first jelly bean is lemon
- B = event that the second jelly bean is strawberry



- Independent: take one jelly bean at random, put it back, and then take a second:

- $P(A) = \frac{10}{15}$ and $P(B) = \frac{5}{15}$

- Dependent: take one jelly bean at random, don't put it back, and then take a second:

- $P(A) = \frac{10}{15}$ and $P(B) = \frac{5}{14}$ if A occurs or $\frac{4}{14}$ if A doesn't occur

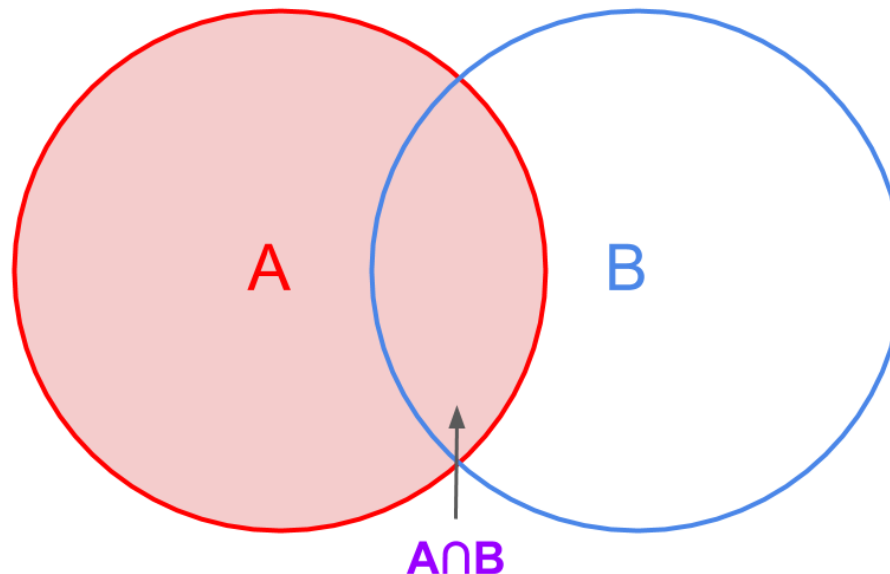
CONDITIONAL PROBABILITY

- The probability of one event occurring given that another event has either occurred or not occurred:

- $P(B|A) = \frac{P(B \cap A)}{P(A)}$

- For independent events, the conditional and unconditional probabilities are the same:

- $P(B|A) = \frac{P(B)P(A)}{P(A)} = P(B)$



CONDITIONAL PROBABILITY: EXAMPLE □

- A = the event that it rains on Monday
- B = the event that it rains on Tuesday
- $P(A) = P(B) = 0.6$ = the probabilities it rains on each day
- $P(A \cap B) = 0.4$ = the probability it rains on Monday and Tuesday
- $P((\text{Not } A) \cap B) = 0.2$ = the probability it doesn't rain on Monday and does rain on Tuesday

If it rains on Monday, then the probability of it raining on Tuesday is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.4}{0.6} = 0.66$$

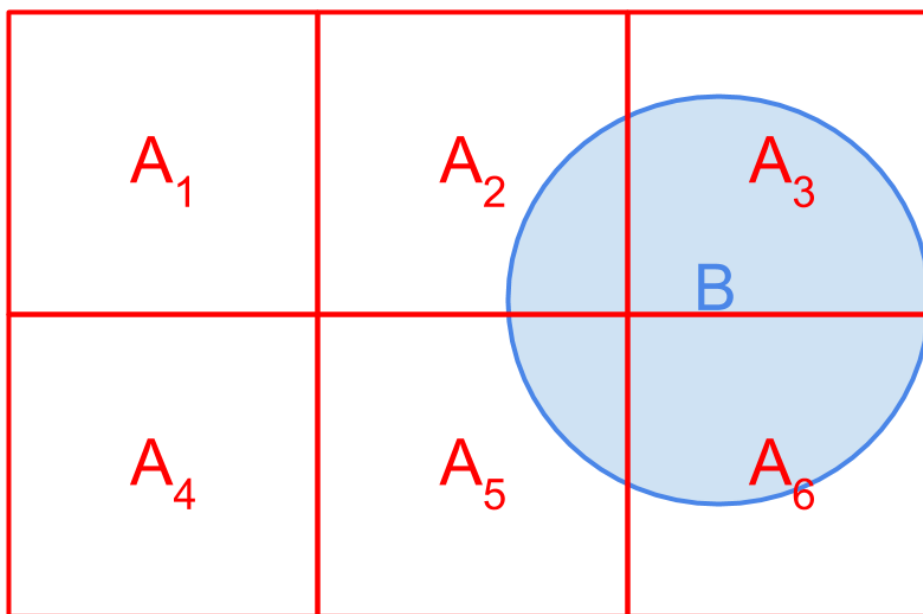
If it doesn't rain on Monday, then the probability of it raining on Tuesday is:

$$P(B|\text{Not } A) = \frac{P((\text{Not } A) \cap B)}{P(\text{Not } A)} = \frac{P((\text{Not } A) \cap B)}{1 - P(A)} = \frac{0.2}{0.4} = 0.5$$

LAW OF TOTAL PROBABILITY

- If one of the events $A_1, A_2, \dots, A_d$ will always occur, then:

$$P(B) = \sum_{i=1}^d P(B \cap A_i) = \sum_{i=1}^d P(B|A_i)P(A_i)$$



BAYES THEOREM

- Bayes Theorem:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

- Proof using the definition of conditional probability:

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

BAYES THEOREM: EXAMPLE □

Suppose a medical test for measles is correct 90% of the time, and that the fraction of the population having measles is 0.001. What is the probability you have measles, given you have received a positive test?

Use Bayes theorem...

$$P(\text{measles}|\text{test positive}) = \frac{P(\text{positive}|\text{measles})P(\text{measles})}{P(\text{positive})}$$

Use the total law of probability...

$$= \frac{P(\text{positive}|\text{measles})P(\text{measles})}{P(\text{positive}|\text{measles})P(\text{measles}) + P(\text{positive}|\text{not measles})P(\text{not measles})}$$

Substitute in values...

$$= \frac{0.9 \times 0.001}{0.9 \times 0.001 + 0.1 \times 0.999} = 0.0089$$

BAYESIAN VERSUS FREQUENTIST INFERENCE

John has misplaced his phone somewhere in the home. He can call his phone and hears it ringing. Problem: which area of John's home should he search?

FREQUENTIST REASONING:

John can hear the phone ringing. He follows the sound to infer the area of his home he must search to locate the phone.

BAYESIAN REASONING:

John can hear the phone ringing. As well as following the sound, John knows the likely locations where he may have misplaced the phone in the past. So, he combines his inferences using the ringing and his *prior* information about the likely locations to identify the area to search.

FROM EVENTS TO RANDOM VARIABLES

Events:

- Receiving a positive measles test
- Rolling a 6 on a die
- It raining on Tuesday

Random variables:

- The result of a measles test: can be positive or negative
- The number on a die: can be 1, 2, 3, 4, 5 or 6
- The weather on Tuesday: can be 'Raining', 'Sunny' or 'Cloudy'

RANDOM VARIABLES

- A random variable X takes values x
- A random variable follows a probability distribution which can be discrete or continuous

Examples of discrete random variables:

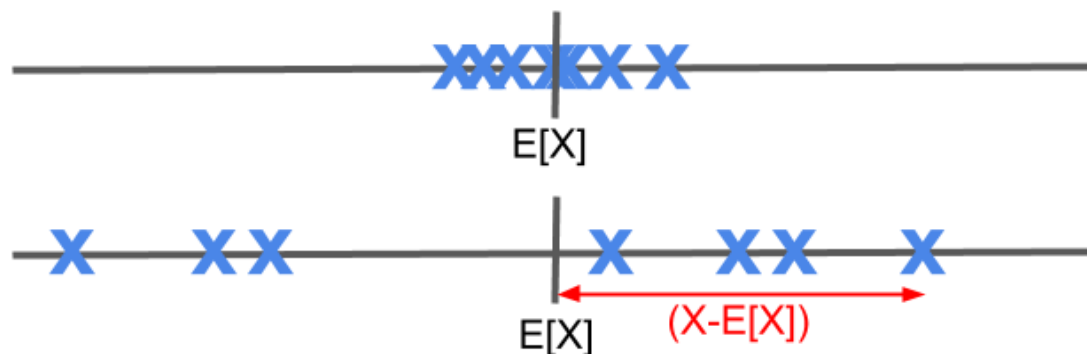
- X is the value shown on a die, x takes values in $\{1, 2, 3, 4, 5, 6\}$
- X is the size of the coffee I will order tomorrow morning, x takes values in {short, tall, grande, venti, trenta}

Examples of continuous random variables:

- X is the height of a person, x takes any positive number
- X is the time (in minutes) it takes to complete a car journey from Birmingham to Edinburgh, x takes any positive number

EXPECTATION AND VARIANCE

- The **expectation** of X is given by $E[X]$
 - The expectation is linear: if a and b are constants, then $E[aX + b] = aE[X] + b$
 - If X, Y are independent random variables, then $E[XY] = E[X]E[Y]$
- The **variance** of X is $\text{Var}(X) = E[(X - E[X])^2]$
 - **Standard deviation**: $s(X) = \sqrt{\text{Var}(X)}$
 - If a and b are constants, then $\text{Var}(aX + b) = a^2 \text{Var}(X)$



MODELLING RANDOM VARIABLES WITH PROBABILITY DISTRIBUTIONS

1. Identify a random variable of interest

- The number of heads we observe when flipping a coin 10 times

2. Collect some data, or use contextual knowledge about the process used to generate the data

- We could do repeat this experiment many times, or use the fact that the probability of observing heads each flip is 0.5

3. Define a formal model which represents how this data could have been generated, to answer questions about the random variable

- Use probability distributions!
- "What is the probability of observing 8 heads out of 10 flips?"
- "What is the expected number of heads we will observe out of 10 flips?"

DISCRETE PROBABILITY DISTRIBUTIONS

- The outcome x can only take values from a **finite (or countable) set**
- Probabilities are defined with a **probability mass function** $P(X = x)$ for each x
- Cumulative distribution function: $F(x) = P(X \leq x) = \sum_{k \leq x} P(X = k)$
- Sum of probabilities is 1: $\sum_k P(X = k) = 1$
- Expectation: $E[X] = \sum_k kP(X = k)$

BERNOULLI DISTRIBUTION

- Two outcomes: $x = 1$ and $x = 0$
- Probability mass function:
 - $P(X = 1) = \text{probability of success} = p$
 - $P(X = 0) = \text{probability of failure} = 1 - p$
- $X \sim \text{Ber}(p)$
- Example:
 - $X \sim \text{Ber}(0.5)$ is the event of seeing heads on a coin flip
 - $E[X] = 0 \times P(X = 0) + 1 \times P(X = 1) = 0.5$

CATEGORICAL DISTRIBUTION

- Random variable X has K possible outcomes: $x_1, x_2, \dots, x_K$
- Probability mass function:

$$P(X = x_k) = p_k \text{ where } \sum_{k=1}^K p_k = 1$$

- $X \sim \text{Categorical}(p_1, p_2, \dots, p_K)$
- Example:
 - X denotes the number shown on the roll of a weighted die
 - $X \sim \text{Categorical}(0.5, 0.1, 0.1, 0.1, 0.1, 0.1)$
 - $E[X] = 1 \times 0.5 + 2 \times 0.1 + \dots + 6 \times 0.1 = 2.5$

BINOMIAL DISTRIBUTION

- Out of a set of n independent Bernoulli trials, each with probability p of a success, the **number of successes** we observe follows a Binomial distribution
- Probability mass function for k in $\{0, 1, 2, \dots, n\}$:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$

- Expectation: $E[X] = np$
- $X \sim \text{Bin}(n, p)$
- Example: X is the number of times we see heads when flipping a fair coin 20 times
 - $X \sim \text{Bin}(20, 0.5)$
 - $E[X] = 20 \times 0.5 = 10$
 - $P(X = 13) = \binom{20}{13} (0.5)^{13} (1 - 0.5)^{20-13}$

$$P(X=13) = 0.07392883300781268$$

MULTINOMIAL DISTRIBUTION

- For n independent trials - each of which leads to a success for exactly one of m categories, with each category having a given fixed success probability - the multinomial distribution models the numbers of successes for each category
- Probability mass function:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m) = \frac{n!}{x_1! x_2! \dots x_m!} p_1^{x_1} p_2^{x_2} \dots p_m^{x_m}$$

- $X_1, X_2, \dots, X_m \sim \text{Multinomial}(p_1, \dots, p_m)$
 - $\sum_{k=1}^m p_k = 1$ and $\sum_{k=1}^m x_k = n$
- Example: rolling the weighted die 20 times and counting the frequency of each value observed
 - X_1 is the number of 1s rolled, X_2 is the number of 2s...

$$P(X_1=10, X_2=2, X_3=1, X_4=2, X_5=1, X_6=4) = 0.0006820094531250049$$

POISSON DISTRIBUTION

- Describes the number of occurrences in a fixed interval where each occurrence is independent of the others and occurs at a rate λ
- Probability mass function:

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

- Expectation: $E[X] = \lambda$
- $X \sim \text{Po}(\lambda)$
- Example: $X \sim \text{Po}(5)$ models the number of customers arriving at a shop in 1 hour, where on average we see 5 customers per hour

CONTINUOUS PROBABILITY DISTRIBUTIONS

- The outcome of an experiment is instead a **continuous** number x
- The probabilities are defined using a **probability density function** $p(x)$
- Area under the whole function is equal to 1:

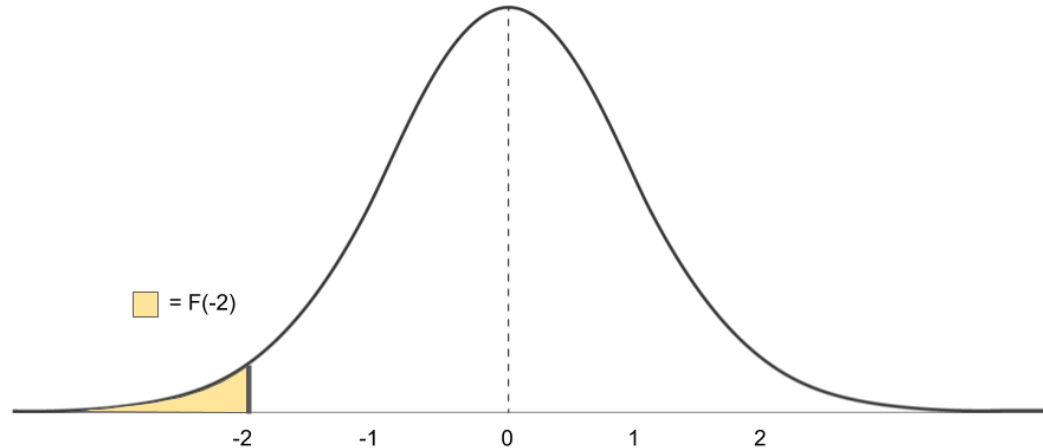
$$\int_{-\infty}^{\infty} p(x)dx = 1$$

- Expectation: $E[X] = \int_{-\infty}^{\infty} xp(x)dx$

CONTINUOUS PROBABILITY DISTRIBUTION: CDF

- Area under the probability distribution function up to a certain value x gives the cumulative distribution function (CDF) of X at x :

$$F(x) = \int_{-\infty}^x p(u) du$$



CONTINUOUS PROBABILITY DISTRIBUTIONS: INTERVALS

- Area under the probability distribution function gives the probability of X being in the interval $[a, b]$:

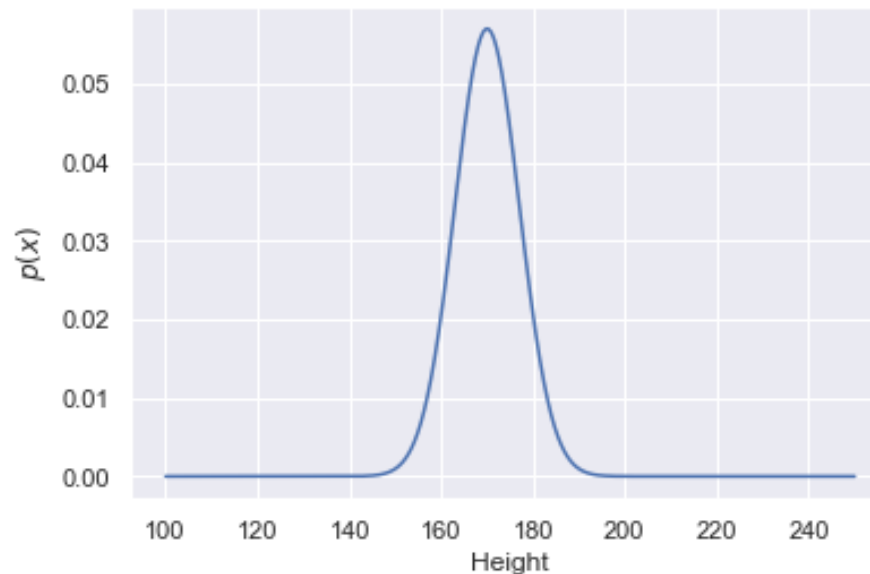
$$P(a \leq X \leq b) = \int_a^b p(x)dx = F(b) - F(a)$$

NORMAL (GAUSSIAN) DISTRIBUTION

- Described by two parameters, the mean μ and variance σ^2
- 95% of the mass lies within 2 standard deviations of the mean
- Probability density function:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- $X \sim N(\mu, \sigma^2)$ generically and $Z \sim \mathcal{N}(0, 1)$ is the standard normal
- Example: $X \sim N(170, 7^2)$ models the height of everybody in this room



EXPONENTIAL DISTRIBUTION

- Describes the time between independent occurrences which occur at rate λ
- Probability density function:

$$p(x) = \lambda e^{-\lambda x}$$

- Cumulative distribution function:

$$F(x) = \int_{-\infty}^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}$$

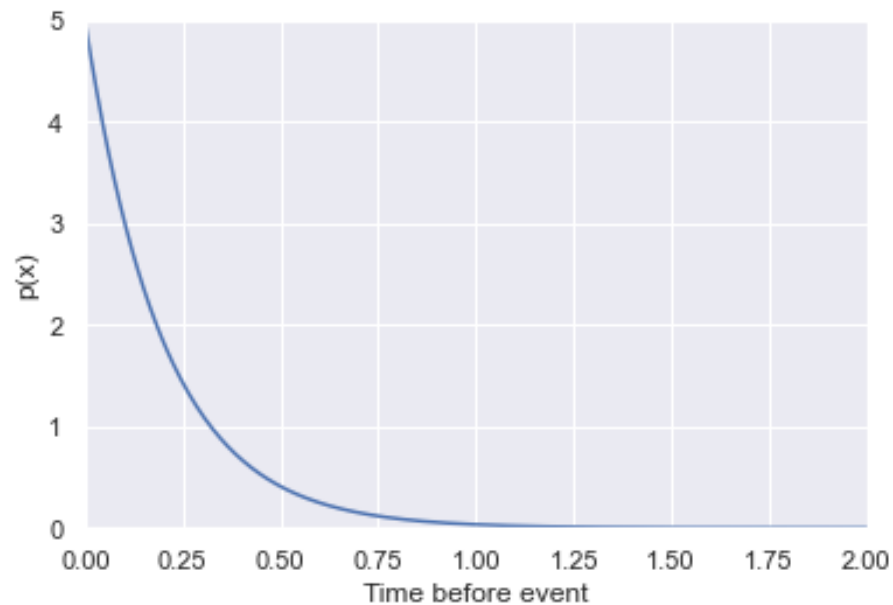
- $X \sim \text{Exp}(\lambda)$

EXPONENTIAL DISTRIBUTION (CONTINUED)

- Probability density function:

$$p(x) = \lambda e^{-\lambda x}$$

- Example: $X \sim \text{Exp}(5)$ is the time between two customers arriving at a shop, which occur at rate 5 per hour

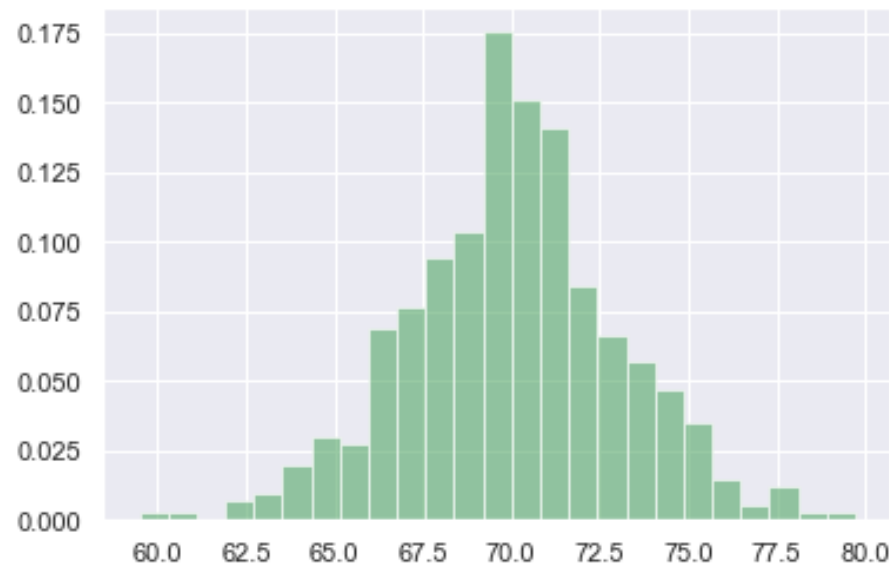


MAXIMUM LIKELIHOOD ESTIMATION: OVERVIEW

- Gaining an intuition about MLE with an example
- Understanding the technical details:
 - Distribution Parameters
 - Likelihood and Log Likelihood
 - Maximising the Likelihood

MLE EXAMPLE

We measure the weights of people in a room in kilograms. The data appear to follow a normal distribution, but which parameters of the normal distribution fit it the best?



□ MLE is a method for estimating population parameters (such as the mean and variance for Normal, rate λ for Poisson, etc.) from sample data such that the probability (likelihood) of obtaining the observed data is maximised.

DISTRIBUTION PARAMETERS

- Distribution parameters index a family of probability distributions.
- A distribution can be fully defined by specifying its parameters.
- Examples:
 - Normal: μ, σ^2 (or σ)
 - Standard normal $Z \sim \mathcal{N}(0, 1)$ so $\mu = 0, \sigma^2 = 1$
 - Poisson and exponential: λ
 - Binomial: n, p

